



Machine Learning | 6037 (20) | 001 (16713)
2026 Fall

Course Information

Modality: In Person

Meeting Day(s): T R

Meeting Time(s): 12:30PM

Meeting Location: WEST

This course meets in Swift 500.

Instructor Information

- **Name:** Anca Ralescu
Instructor Title: Professor
Instructor Qualifications: PhD
Email: ralescal@ucmail.uc.edu

Instructor Contact Information

Virtual Office Hours

will be held on Zoom by appointment

Office Location: 310 Mantei Bldg

Office Hours: by appointment

Office Hours Location: 310 Mantei Bldg.
Email: ralescal@ucmail.uc.edu

Teaching Assistants and/or Peer Educators

Name: Anca Ralescu
Email: ralescal@ucmail.uc.edu

Name: TA will be announced after assignment by the CS Department Head
Email: TBA

Course Description

The goal of this course is to introduce students to the field of Machine Learning. The course covers traditional machine learning algorithms and their implementations along with discussions of concrete problems where these algorithms are suitable. Topics covered by course include: Concept Learning and the General-to-Specific Ordering Decision Tree Learning Artificial Neural Networks Evaluating Hypotheses/Bayesian Learning Computational Learning Theory Instance-Based Learning Genetic Algorithms Learning Sets of Rules Analytical Learning Combining Inductive and Analytical Learning Reinforcement Learning. - Prerequisite Definition: To take this course you must: Be enrolled in one of these Programs 20DOC 20MAS.

Course Overview

Machine Learning is a continuously, rapidly evolving domain of intelligent computing. It can be described broadly as models and algorithms to recognize and compute with patterns. This course will explore Machine Learning from the most basic to more advanced models. Students will implement the corresponding algorithms. These are the basis for understanding the newest developments in this field.

Course Format

This is a lecture based course, which meets twice a week for one hour and 20 minutes. Students will be assigned homework assignments which consist in

implementation of the algorithms discussed. Two exams will be given. The format of the exam will most likely be online. Tentative dates will be listed in the weekly schedule.

Course Learning Outcomes

Illustrate the development of the field of machine learning the various algorithms that form the core of this discipline along with their implementations and practical applications. ABET Student Learning Outcome: Integrate knowledge of computing and mathematics appropriately in computer science. ABET Student Learning Outcome: illustrate mathematical foundations algorithmic principles & CS theory in modeling and design of computer-based systems demonstrating comprehension of the tradeoffs involved in design choices. ABET Student Learning Outcome: Select mathematical foundations and database principles to model information.

Course Related Costs

Students are not required to purchase additional resources for the course.

Required Course Materials

[University of Cincinnati Bookstore](#)

- **Title: This class has no required readings or materials**

Artificial Intelligence (AI) Guidelines

Not appropriate to use – In this course, it is expected that all submitted work is produced by the students themselves, whether individually or collaboratively with other students (in accordance with the class policies). The use of Generative AI tools is not aligned with the goals of this course.

What coursework can be completed with the help of generative AI tools?

AI is not permitted - Students must not seek the assistance of Generative AI Tools like ChatGPT. Students' use of a Generative AI Tool to complete an assignment in any capacity constitutes academic dishonesty.

Course Calendar

- Week/Module: 1: 8/25 & 27
Topic: A FEW SIMPLE (EARLY) CONCEPT LEARNING ALGORITHMS inspired from AI: FindS, Candidate Elimination, Version Space
Material: Instructor Notes & Slides provided on Canvas
- Week/Module: 2: 9/1 & 3
Topic: Intro to ML: Overview, Regression
Material: Instructor Notes & Slides provided on Canvas
- Week/Module: 3: 9/8 & 10
Topic: Regression Continued : Overfitting, Generalization, Bias-Variance tradeoff
Material: Instructor Notes & Slides provided on Canvas
- Week/Module: 4: 9/15 & 17
Topic: Regression Continued : Logistic Regression , LASSO, Elastic Net
Material: Instructor Notes & Slides provided on Canvas
- Week/Module: 5: 9/22 & 24
Topic: Decision Trees (Part 1)
Material: Instructor Notes & Slides provided on Canvas
- Week/Module: 6: 9/29 & 10/1
Topic: Decision Trees (Part 2); Ensemble Learning, Random Forests , Classifier Evaluation: Accuracy, Precision, Recall, F-measure; Confusion Matrix, ROC Curve
Material: Instructor Notes & Slides provided on Canvas
- Week/Module: 7: 10/6 & 8
Topic: Bayesian Learning
Material: Instructor Notes & Slides provided on Canvas
- Week/Module: 8: 10/13 & 15
Topic: The Expectation Maximization Algorithm
Material: Instructor Notes & Slides provided on Canvas
- Week/Module: 9: 10/20 & 22

Topic: Reinforcement Learning; Markov Decision Processes
Material: Instructor Notes & Slides provided on Canvas

- Week/Module: 10: 10/27 & 29
Topic: Neural Networks (Part 1)
Material: Instructor Notes & Slides provided on Canvas
- Week/Module: 11: 11/3 & 5
Topic: Neural Networks (Part 2); Learning a Bayesian Network
Material: Instructor Notes & Slides provided on Canvas
- Week/Module: 12: 11/10 & 12
Topic: Probably Almost Correct (PAC) Learning; Empirical & Theoretical Risk, VC dimension, Kernels
Material: Instructor Notes & Slides provided on Canvas
- Week/Module: 13: 11/17 & 19
Topic: • Support Vector Machines (SVM); Clustering: Kernel Based Clustering, Cluster Validity
Material: Instructor Notes & Slides provided on Canvas
- Week/Module: 14: 11/24 & 26
Topic: Dimensionality Reduction; Computing Projections; Random Projections
Material: Instructor Notes & Slides provided on Canvas
- Week/Module: 15: 12/1 & 3
Topic: Catch All week
Material: Instructor Notes & Slides provided on Canvas

Assessment/Activities and Grading Policy

There will be approximately 7 assignments. Assignments are to be completed in MATLAB (preferred) or Python. No libraries or AI are to be used in completing the programming assignments. Each algorithm must be written from scratch.

1. For the homework assignments, students are required to work in teams of NO MORE than 3 students per team.
 2. If a student/group selects one language, they must use that language for the whole semester.
 3. Changes of teams are permitted ONLY ONCE during the semester.
- Up to 7 quizzes if possible
 - There will be two exams
 1. First Exam: approximately midway through the semester (E)
 2. Second Exam: last day of class (E)

Grading

- The final grade for the course is based on a weighted average score obtained at the homework assignments, quizzes, and exams.
- All the homework assignments together count as one exam(E)
- All the quizzes together count as half of an exam. (0.5E)
 1. $\text{SCORE} = (E + E + E + 0.5E)/3.5$
 2. Let M be the highest score obtained in the class. That is:
 3. $M = \max \{S \mid \text{where SCORE designates all the scores in the class}\}$

Letter grades are assigned based on the percentage of M, as shown in the table below.

Tentative Weighting	
Assignment	Weighting
Homework Average	28%
Quizzes Average	16%
Exam 1	28%
Exam 2	28%
Total	100%

Final letter grades for the course will be determined using your final numerical grade and the following ranges. Please note that the instructor reserves the right to modify the range for each letter grade. However, the ranges posted here provide the minimum score you must achieve to guarantee the grade indicated.

Numeric Range (brackets are wrt M, the highest class score)	
Grade	Range
A	92-100
A-	89-92
B+	86-89
B	82-86
B-	79-82
C+	76-79
C	72-76
C-	69-72
D+	66-69
D	62-66
D-	59-62
F	Below 59

Test Proctoring

To be announced.

Pass/Fail, Audit, and Withdrawal

- **Students enrolled as Audit must complete at least 50% of the course work.**
- For information on Adding, Dropping, and Withdrawing from classes, please visit: [University of Cincinnati Office of Registrar](#)
- For information on Grading Scales and Definitions, please visit: [University of Cincinnati Grading Scales and Definitions](#)

Communication Policy

Communication from the instructor will typically be through Canvas (Announcements or email). Students are encouraged to use Canvas to communicate with the instructor and/or the TAs.

Technology use during/for class

For technology resources such as the DTS Knowledge Base (KB) tutorial articles and the Help Desk, visit the [DTS Help Page](#). Currently the student text is blank.

Attendance Policy

Students at the University of Cincinnati are expected to attend classes in order to meet the learning objectives for the course. Students are expected to follow the class attendance policy as outlined by the instructor. Students participating in official University sponsored activities, where the student is representing the University, will meet with instructor related to absences due to university-sponsored activities. When such absences occur, students need to provide documentation of the event(s), and work with the professor to develop a written plan for completion of missed assignments or other course requirements with as

much advance notice as possible. Any absence due to official University business may still count against the total number of allowable absences.

Faculty Attendance

There is a 10 minute interval between this class and the class before it which is taught in a different building. Although all effort will be made by the instructor to be on time, it is possible that the instructor will be a few minutes late.

Class Cancellation Policy

In the rare case that a class must be cancelled, or that a faculty member is absent, faculty will post an announcement on Canvas or email information to students. Faculty will attempt to communicate class cancelations with as much advance notice as possible. Students should be sure that their LMS email is current and valid to ensure emails are received.

For information on class cancellation due to emergency procedures follow the link below:

[Emergency Closing Procedure.](#)

Notice of Non-Discrimination

- For the University of Cincinnati's Notice of Non-Discrimination, please visit: [UC Notice of Non-Discrimination](#)

Student Religious Accommodations

- For information on the University's Religious Accommodations Policy, please visit: [Student Religious Accommodations Policy](#)

Accessibility Policy

If you have a physical, psychological, or cognitive disability (learning, ADD/

ADHD, psychological, visual, hearing, physical, cognitive, medical condition, etc.), UC's Accessibility Resources and your college's Access Coordinator are here to ensure you receive reasonable accommodations.

I encourage you to contact [Accessibility Resources](#) to schedule a confidential meeting to discuss services and accommodations, or to begin by registering for accommodations. UC uses an online system called [UC Accommodate](#), where you'll be able to register and manage your accommodations (use your UC username and password for access).

Below are a few helpful UC Accommodate resources:

[UC Accommodate Student Portal Overview Links to an external site.](#)

[How to Schedule an Appointment in UC Accommodate Links to an external site.](#)

[How to Submit a Semester Request in UC Accommodate](#)

Academic Integrity and Learning Commons

For information on the University's Academic Misconduct Policies.

[Academic Misconduct | University of Cincinnati](#)

[Learning Commons](#)

The Learning Commons provides centralized academic support for all University of Cincinnati students by offering tutoring, coaching, writing support, and many additional programs that you can access via registered courses, appointments, or just dropping into the office. Learning Commons professionals and Peer Educators are available to help create flexible academic success programming, whether you need some assistance with brainstorming ideas for your assignment, improving a draft, or managing your time. Check out the Learning Commons website to learn about available support and how to schedule appointments. You can also call the Learning Commons at 513-556-3244 or stop by the desk on the ground floor of French Hall West or Langsam Library at the Academic Writing Center (AWC).

Ethics and integrity are core values that should guide our conduct and decisions

as members of the UC community and as engineering and technology professionals. Course assignments and tests are designed to have educational value and prepare you to be a competent and ethical professional. Therefore, material presented by you to satisfy course requirements is expected to be the result of your own original scholarly efforts, unless collaboration is explicitly allowed on a particular assignment.

University Rules, including the [Student Code of Conduct](#) and other documented policies of the department, college, and university related to academic integrity, will be enforced. Any violation of these regulations, including acts of plagiarism or cheating, will be dealt with on an individual basis according to the severity of the misconduct and can result in severe consequences, including potential dismissal from the college after the second offense.

Title IX

For information on the University's Title IX Policy click the link:

[Title IX Policy - Ethics, Compliance & Community Impact | University of Cincinnati](#)

Help for Student Victims & Student Survivors

The [Help for Student Victims & Student Survivors website](#) provides resources for victims or survivors of sexual harassment, including sexual assault, dating or domestic violence, gender-based harassment, or stalking. Staff are available 24-hours a day for confidential advice and assistance.

Bearcats Pantry and Resource Center

[The Bearcats Pantry and Resource Center](#) is available to all UC students. From free food to social services support, they provide a range of programs, services, and supplies. Each student may visit once per week and receive up to 20lbs of goods per visit. You will need to make a [PantrySOFT account](#) and present your physical Bearcat Card for entry. They also encourage you to bring a reusable bag when you visit.

Mental Health Resources

Mental Health counseling and a variety of resources are available to UC students through [Counseling & Psychological Services \(CAPS\)](#). The Crisis, Assessment, Referral, Evaluation Team (“CARE Team”) responds to reports about students whose behavior is raising concerns within the University community. More information about the CARE Team can be found in the [University's Mental Health Assessment Policy](#).

UC Public Safety

The Department of Public Safety at the University of Cincinnati is committed to providing a safe campus environment for students, faculty, staff and visitors. Visit the [UC Public Safety website](#) to learn more about public safety and emergency response resources and services.

Inclusivity

This class is committed to the fundamental principles of academic freedom and human dignity. Diversity in all forms is something we welcome, we foster, and we prize. We believe that honest attempts to understand the perspectives of others facilitates learning, and we will strive to achieve this goal at all times. We strongly disavow discrimination -- including harassment -- on the basis of race, national or ethnic origin, religion, sex or gender identity, disability, age, sexual orientation, or veteran status. We expect that each of us will hold one another accountable for maintaining these ideals.

All are welcome and considered a valuable addition to the university community. You should consider my classroom as an inclusive and safe space to express your ideas and viewpoints. No discrimination is accepted or tolerated in this course. It is the goal for you to be successful and to thrive to your highest potential. [UC Notice of Non-Discrimination](#)

Special Statement on COVID-19

[UC's COVID-19 Public Health Webpage](#) contains the most up-to-date policies and practices regarding COVID-19 precautions at UC. Remember, sick and/or quarantined students should not come to class. Instead, please contact the instructor to discuss appropriate online accommodations.

Right to Modify the Syllabus

I reserve the right to update this syllabus as class needs arise. Be assured that I will communicate to you any changes to our schedule, syllabus or policies quickly and efficiently through **Canvas Announcements, and/or email.**